Haiyang Wang

Applied Mathematics Ph.D. Researcher | Machine Learning • Information Theory • Scientific Computing
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Website | GitHub | LinkedIn | Google Scholar

Education

Yale University — **New Haven, CT**
Ph.D. in Applied Mathematics, expected May 2027 — *Aug 2023–Present*
Advisors: Yihong Wu and Ronald Coifman

New York University — **New York, NY**
B.A. in Mathematics and B.A. in Computer Science — *Aug 2018–Dec 2022*

Honors & Awards

IEEE Jack Keil Wolf ISIT Student Paper Award — 1 of 3 winners among 400+ eligible papers — **2026**
Related travel support: ISIT Student Travel Grant; Yale GSA Conference Travel Fellowship — **2026**

Research Experience

Yale University — **New Haven, CT**
Ph.D. Researcher — *Aug 2023–Present*

Machine Learning Theory for XGBoost *with Yihong Wu*

- Derived convergence rates for XGBoost in fundamental settings, translating theoretical guarantees into principled parameter selection.
- Characterized consistency under missing covariates and quantified the approximation error when consistency fails.

Amplitude-Constrained Gaussian Channels *with Luca Barletta and Alex Dytso*

- Developed an approximation-theoretic framework that converts optimal-distribution questions into best approximation of a target law on a compact domain; this insight has since been applied to other optimal-distribution problems.
- Used the framework to prove a new $\Omega(A\sqrt{\log A})$ lower bound on the capacity-achieving input support size, improving the prior linear bound and ruling out a longstanding linear-scaling conjecture.

Flatiron Institute, Simons Foundation — **New York, NY**
Research Analyst, managed by Leslie Greengard and Manas Rachh — *Jan 2023–Jul 2023*

- Developed a high-order scattering-matrix method for Stokes flow in large-scale channel geometries, achieving a $100\times$ speedup.
- Designed fast solvers for Helmholtz, Lippmann–Schwinger, and Stokes equations; contributed to the chunkIE MATLAB integral-equation toolbox.

Selected Journal Publications

Haiyang Wang, Luca Barletta, and Alex Dytso. "An improved lower bound on cardinality of support of the amplitude-constrained AWGN channel." *IEEE Transactions on Information Theory*, 2026. **2026 Jack Keil Wolf ISIT Student Paper Award**.

Haiyang Wang, Fredrik Fryklund, Samuel Potter, and Leslie Greengard. "Scattering theory for Stokes flow in complex branched structures." *Journal of Computational Physics*, article 115148, 2026.

Technical Skills

Programming: Python (pandas, NumPy, SciPy, JAX, PyTorch)
Methods: Machine learning theory, statistical learning, gradient-boosted trees, deep learning, numerical PDEs, boundary integral methods